

Functional Features

Starting Date	08 October 2026
Team ID	UNM1611PUNITHA
Project Name	AI STUDY BUDDY API

Instructions

List the functional capabilities included in the implemented solution. Feature descriptions and counts below are derived from the supplied source code.

Functional Feature Summary

#	Feature	Implementation / outcome
1	User registration	Creates student/admin accounts; rejects duplicate email; hashes passwords with bcryptjs.
2	User login	Validates credentials and returns access and refresh JWTs.
3	Token refresh	Validates refresh JWT from x-refresh-token and issues new tokens.
4	Logout	Returns logout confirmation; token disposal is handled client-side.
5	Protected API access	JWT Bearer authentication protects material and admin resources.
6	Role-based access	Admin-only middleware restricts /api/admin operations.
7	Study-material upload	Accepts .txt, .md and .pdf extensions through Multer; 5 MB limit.
8	Material listing	Lists the authenticated user's materials; admin can list all.
9	Material retrieval	Returns a single material while enforcing student ownership.
10	Material deletion	Deletes a material after ownership/admin authorization.
11	AI summarization	Gemini generates a concise bullet-point summary and stores it.
12	AI flashcards	Gemini generates JSON flashcards with question/answer pairs.
13	AI MCQ quiz	Gemini generates JSON multiple-choice questions with options and answers.
14	Personalized study plan	Gemini creates a day-by-day plan from goal, hours/day and duration.
15	Admin user management	Admin can list users and delete a user with their materials.
16	Admin statistics	Reports total student users and total stored materials.
17	Health endpoint	Root endpoint reports that the AI StudyBuddy API is running.

API Feature Map

Module	Endpoint	Purpose
Auth	POST /api/auth/register	Register a user
Auth	POST /api/auth/login	Authenticate user
Auth	POST /api/auth/refresh	Refresh JWT pair
Auth	POST /api/auth/logout	Return logout response
Materials	POST /api/materials/upload	Upload study material
Materials	GET /api/materials	List materials
Materials	GET /api/materials/:id	Get one material
Materials	DELETE /api/materials/:id	Delete material
AI	POST /api/materials/:id/summarize	Generate summary
AI	POST /api/materials/:id/flashcards	Generate flashcards
AI	POST /api/materials/:id/quiz	Generate MCQ quiz
AI	POST /api/materials/:id/study-plan	Generate study plan
Admin	GET /api/admin/users	List users
Admin	DELETE /api/admin/users/:id	Delete user and materials
Admin	GET /api/admin/stats	Get user/material counts

Functional Scope Note

The supplied project is an API/backend implementation. A separate frontend, automated API test suite, background job queue, token revocation store, or production deployment configuration is not present in the ZIP and is not counted as an implemented feature.